

Sebastian Sotomayor

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Education

University of Central Florida

Bachelor of Science in Computer Engineering / GPA: 3.505

Orlando, FL

Expected May 2026

Relevant Coursework: Data Structures and Algorithms I & II, Computer Architecture, Electronics I, Embedded Systems

Technical Skills

Programming & Software: C, Embedded C, Java, C++, Python, MIPS, HTML/CSS, JavaScript

Hardware & Tools: PCB Design, FPGA Programming, Oscilloscope, Waveform Generator, Multimeter

Languages: Fluent in English and Spanish

Professional Experience

General Dynamics Mission Systems (GDMS)

Orlando, FL

Embedded Software Engineering Intern

September 2025 – Present

- Implement and optimize C/C++ code for real-time operating systems (RTOS) to improve reliability and execution performance.
- Assist in hardware–software integration, troubleshooting communication protocols (UART, SPI, I2C, BLE), and validating system-level functionality.
- Utilize Git, debugging tools, and embedded workflows to support efficient collaboration and system testing.

Winter Park Tennis Center

Winter Park, FL

Maintenance Worker

May 2024 – May 2026

- Maintain and repair court surfaces, nets, and fencing to ensure a safe playing environment.
- Performed inspections and sanitation to improve facility readiness and patron satisfaction.

Projects and Organizations

Multi-Adjustable Port Station

Orlando, FL

Capstone Project

August 2025 – April 2026

- Developed a real-time smart charging station using an ESP32, Raspberry Pi, and full-stack web interface to monitor and control multiple charging ports.
- Integrated FreeRTOS firmware, UART communication, live telemetry, configurable current thresholds, timer-based cutoff logic, and fault detection.
- Collaborated with a multidisciplinary team to validate hardware/software integration and present the project as a UCF CECS Senior Design Showcase finalist.

Institute of Electrical Electronics Engineers (IEEE)

Orlando, FL

SoutheastCon Hardware Competition

September 2024 – March 2025

- Designed and integrated power electronics systems for an autonomous robot, ensuring stable power delivery through buck converters and voltage dividers.
- Managed battery configurations and optimized electrical efficiency, contributing to reliable robot performance and competitive success.

University of Central Florida – Embedded Systems

Orlando, FL

Embedded Systems Laboratory

January 2025 – April 2025

- Completed 11 projects using the MSP430FR6989, integrating peripherals such as LCDs, joysticks, and optical sensors via SPI, I2C, and UART.
- Developed interrupt-driven firmware in Embedded C, leveraging timers, ADCs, and low-power modes for real-time operation in constrained environments.